Gas Laws Practicals

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1 Practical 1: Boyle's law

1.1 Aim

To Investigate Boyle's Law

1.2 Variables

Independent: Volume(Length)

Dependent: Pressure

Control: Mass, Temperature

1.3 Method

1. Given measuring the cross sectional area of the tube is rather difficult; the area stays constant throughout, therefore the length of the fluid in the tube is proportional to the volume of the fluid in the capillary tube.
2. Set up the apparatus shown in the diagram, ensuring the tube is empty
3. Drawing in air each time to the tube (slowly), record the height (L_n /cm), until the total volume of the tube is filled up
4. Repeat the experiment removing the water in the tube and record the mean volume for each mass ($\overline{L_n}$ /cm)
5. Plot a graph of P / Pa (y-axis) against $\frac{1}{L}$ / cm^{-1} (x-axis), drawing a best-fit line, and deduce from the straight line through the origin that $P \propto \frac{1}{L}$.

1.4 Risk assessment

Risk	Hazard	Mitigation(s)

1.5 Results

L / cm	$1/L$ / cm^{-1}	P / Pa
50.2	0.0199	1.10
47.8	0.0209	1.15
46.7	0.0214	1.20
39.2	0.0255	1.40
35.7	0.0280	1.55
28.4	0.0352	1.93
20.7	0.0483	2.60

Table 1: Results table.

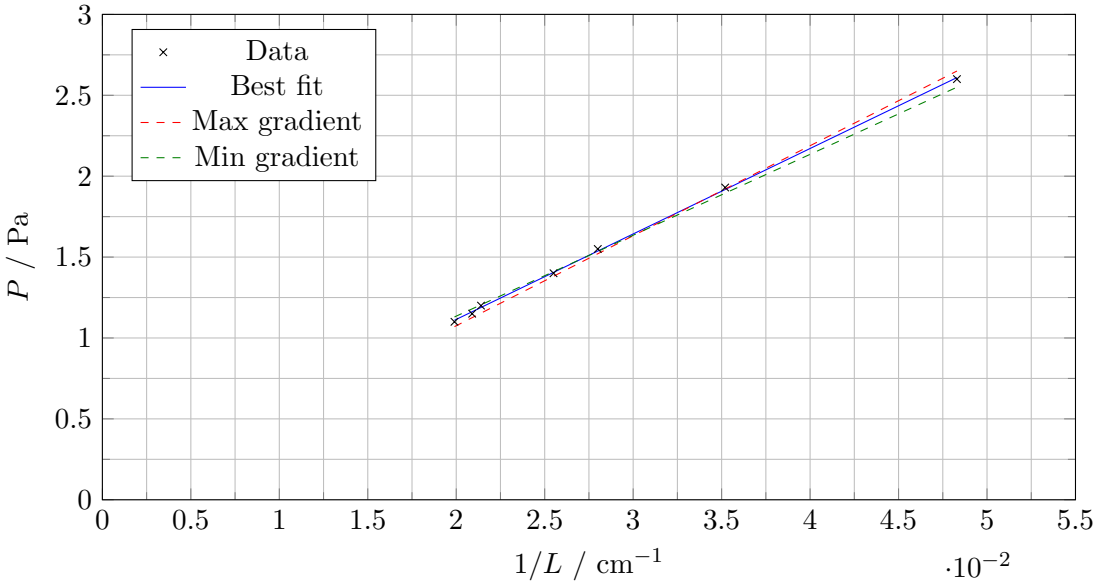


Figure 1: Graph of P against $\frac{1}{L}$ with best-fit line and worst-fit lines.

1.6 Analysis

Gradient of maximum line: 55.63
Gradient of minimum line: 50.00
Mean gradient: 52.82
Uncertainty in gradient: ± 2.82

$$P = k \cdot \frac{1}{L} \quad \Rightarrow \quad k = \text{constant} \tag{1}$$

1.7 Uncertainties

1.8 Evaluation

2 Practical 2: Charles' law

2.1 Aim

To Investigate Charles' Law

2.2 Variables

Independent: Temperature

Dependent: Length(Volume)

Control: Mass, Pressure

2.3 Introduction

Measuring the cross-sectional area of the capillary tube is difficult. Since the area stays constant throughout, the length of the fluid in the capillary tube is proportional to the volume of the fluid in the capillary tube:

$$L \propto V \tag{2}$$

2.4 Method

1. Set up the apparatus as shown in the diagram, ensuring the hot water in the beaker or measuring cylinder fully submerges the air sample.
2. Stir the water well using the thermometer, then read and record the initial temperature (θ_i / °C) and the length of the air sample (L / cm).
3. Add ice and stir well, taking measurements every 5 °C down to room temperature.
4. Plot a graph of θ / °C (x-axis) against L / cm (y-axis), drawing a best-fit line.
5. Deduce the point of absolute zero, θ_0, from the graph by extrapolating the line to the x-intercept (where $L = 0$).

2.5 Risk assessment

Risk	Hazard	Mitigation(s)
Boiling water	Burns	Handle boiling water with care, taking caution not to cause any spills.

2.6 Results

θ / °C	L / cm
93.0	16.3
90.5	15.8
89.0	15.5
86.0	15.0
84.0	14.6
83.0	14.3
81.0	13.9
70.0	13.5
65.0	12.7
62.5	12.5
60.0	12.0
52.5	11.6

Table 2: Results table.

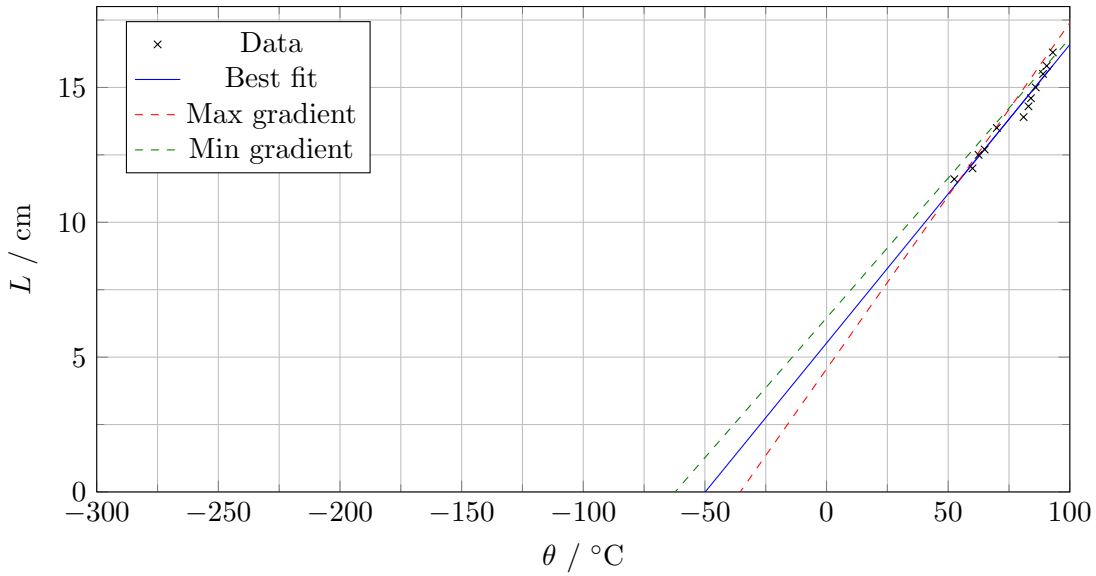


Figure 2: Graph of L against θ

2.7 Analysis

Gradient of maximum line: 0.128
Gradient of minimum line: 0.104
Mean gradient: 0.116
Uncertainty in gradient: ± 0.012

$\theta_0 = -49 \pm 13\,°\mathrm{C}$

$$V \propto T, \qquad T\,/\,\mathrm{K} = \theta\,/\,°\mathrm{C} + 273.15 \tag{3}$$

2.8 Uncertainties

2.9 Evaluation

References

https://pmt.physicsandmathstutor.com/download/Physics/A-level/Notes/AQA/Practical-Skills/RP%2008%20-%20Boyle's%20and%20Charles's%20Laws.pdf (accessed 25/09/2026, 18:58)